



The identification of sub-centres in two Italian metropolitan areas: A functional approach[☆]

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ARTICLE INFO

Article history:

Received 28 July 2011

Received in revised form 11 April 2012

Accepted 24 April 2012

Available online 20 May 2012

Keywords:

Sub-centres

Polycentricity

Metropolitan areas

ABSTRACT

This work introduces a method aimed at identifying metropolitan sub-centres based on an interaction approach, using commuting flows. In addition, it compares the results with other approaches based on employment and/or density thresholds commonly used in the literature. Although the scientific literature and the policy debate on polycentricity are increasingly devoted to its functional dimension, empirical research aimed at identifying metropolitan sub-centres mainly relies on employment density, which refers to a more morphological approach. After discussing the concept of centre and the contemporary literature on its empirical identification, the proposed methodology is applied to the metropolitan areas of Rome and Milan. Such a method shows a better fit compared to other approaches based on job density. Results are found to be remarkably sensitive to the method applied and are dependent, to some extent, on the different concept of centre that is adopted in the characterisation of metropolitan spatial structures.

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Introduction

Contemporary Metropolitan Areas (MAs) are characterised by ever more complex spatial structures that are increasingly different from the archetype of the monocentric city. With a particular intensity in the last two decades, metropolitan employment has been scattered throughout the urban territory, changing the traditional patterns of urban spatial organisation towards dispersed or polycentric structures (García-López and Muñiz, 2010). In addition, agglomeration externalities – whose role in economic growth and productivity is increasingly studied – have extended geographically beyond the boundaries of single urban cores and are increasingly shared among different interconnected centres (Meijers & Burger, 2010). Polycentricity is also thought of as a normative tool to reach important objectives in terms of environmental sustainability and social cohesion (Davoudi, 2003). As a matter of fact, measuring polycentricity – and, more generally, urban spatial structure – has become an important research topic.

A necessary first step in the characterisation of polycentric MA concerns the identification of metropolitan sub-centres (Anas, Arnott, & Small, 1998), though this can be thought of as an independent topic with its own relevance. In fact, the identification of sub-

centres allows a wider knowledge of metropolitan spatial organisation, which is necessary for any spatial planning policy. The identification of metropolitan sub-centres also allows the recognition of places where the priorities of public investments are assigned, places that shape the agglomeration effects within their region. In fact, if we accept that external economies of agglomeration have regionalised their space of influence (Richardson, 1995), then polycentric urban structures could have a role in enhancing this source of economic advantages. Only by providing reliable tools to measure urban spatial structure it would be possible to verify the effects of polycentricity on metropolitan economic performance, a topic which is still empirically understudied.

The aim of this work is to propose a methodological approach to identifying metropolitan sub-centres. For this purpose, two Italian MAs – Rome and Milan – are chosen as study areas. The proposed methodology is based on an interaction and functional approach and results are compared with those obtained from two other methods based on density and employment size. This work aims at filling a gap in the literature on the description of urban spatial structure. In fact, while increasing attention has been devoted to functional relations that occur among territorial nodes within metropolitan areas – or even at higher spatial levels (Derudder, Taylor, Witlox, & Catalano, 2003) – much less attention has been devoted at identifying metropolitan centralities using interaction measures in place of density-based approaches. Comparative studies between the results of the two approaches are also lacking.

The works aimed at identifying employment sub-centres that adopt morphological and density-based methodologies are based on the bid-rent theoretical tradition (Alonso, 1964; Mills, 1972;

[☆] The views expressed herein are those of the author and do not necessarily reflect the views of the OECD.

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Muth, 1969). The latter approaches are particularly appropriate when new sub-centres emerge as a consequence of the processes of employment decentralisation from a Central Business District (CBD) to the hinterland due to high congestion and land prices. Hence, centrifugal and centripetal forces are at the base of the changing metropolitan spatial structure, which evolve from a monocentric model to a polycentric or a dispersed one (Lee, 2007). However, the procedure of sub-centre identification proposed in this work is more closely connected with the Central Place theoretical tradition (Christaller, 1966; Lösch, 1954), which is more focused on the hierarchical relations across multiple centres located within a given urban system, that in the case of this paper are metropolitan areas. In central place theory, the hierarchical rank of each node determines its centrality, which in turn depends on the functions that each centre provides. In this work, travel-to-work flows are used as a measure of relationship between territorial nodes, since central places are expected to have an excess of labour demand compared with non-central places. Underlying this perspective, there is a particular meaning of the concept of centre which depends on functional relations among nodes. In fact, the importance of a centre is connected with the excess of functions it supplies relative to those demanded by local inhabitants (Preston, 1971). Thus, the importance of central places should be assessed by looking at the strength of their relations rather than on their internal characteristics (Burger & Meijers, 2012).

The functional approach to polycentricity can be particularly sound for the study of European – and especially for Italian – MAs. In fact, many European MAs emerged mostly as a result of an integration or coalescence process between old and pre-existing cities (Calafati, 2002; Champion, 2001), rather than a decentralisation process from a single and congested urban core. Milan and Rome represent two paradigmatic examples of metropolitan areas in which territorial coalescence brought about a complex spatial structure in which previously existing urban nodes integrated in a larger area where some nodes gained importance and some others lost.

The identification of metropolitan sub-centres is carried out in this work by analysing their functional characteristics rather than their morphological features. Thus, in order to identify metropolitan sub-centres, commuting flows between urban nodes (municipalities) and the diversity/completeness of their sectorial structures have been analysed. These aspects allow us to draw a hierarchic ranking between places and to identify those municipalities that turn out to be more locally central. On the other hand, restricting the analysis to population and density patterns within MAs is expected to catch another dimension of spatial structure. Such a dimension is founded on the absolute importance of a centre (global nodality), with less focus on its role for the surrounding territory (Lukermann, 1966).

The remainder of the paper is organised as follows. In the next section, a review of the literature on sub-centre identification is carried out, distinguishing the major static and morphological methodologies from interaction-based ones. The third section explains the procedure that has been used in this work to identify metropolitan sub-centres, while the fourth section presents the results of its application for Rome and Milan MAs. Results are also compared with those that use other more traditional methods found in the literature. A comparison is discussed in the fifth section. Finally, some concluding remarks are provided in the last section.

Literature review

Static methodologies

The polycentric structure at the intra-urban level, and the possibility that multiple centres can exist within urban areas,

was already clear in the first half of the 20th Century (Harris & Ullman, 1945). However, the literature on the identification of sub-centres has grown in a particularly rapid way in the last twenty years, improving its degree of rigorousness and adaptability to different territorial contexts. The most commonly employed methodologies to identify sub-centres make use of density measures. This is because of the meaning that is given to the concept of centre (or sub-centre). In fact, a metropolitan centre could be thought of as a place characterised by a high concentration of employment and activities with respect to the surrounding territory. This definition works particularly well for MAs that grow and evolve mainly by decentralisation from a congested CBD towards new concentrations of activities (sub-centres) due to a trade-off between economies and diseconomies of agglomeration.

In a chronological order, one of the first straightforward methods of identification is that by Giuliano and Small (1991), which is based on employment and density thresholds. The two authors identify 32 sub-centres in the Los Angeles MA by selecting those transportation analysis zones (TAZs) that show a gross employment density of at least 10 jobs per acre and a total employment of at least 10,000 units. The same method – but with different cut-off values – has been adopted by many other authors, such as Small and Song (1994), Bogart and Ferry (1999) and Cervero and Wu (1998). They analyse the MAs of Los Angeles, Cleveland and San Francisco, respectively. More recently, Muñiz, García López, and Galindo (2008) used relative thresholds for the MA of Barcelona, selecting as sub-centres those municipalities with a gross employment density higher than the MA average that account at the same time for at least 1% of the total metropolitan employment. Since these methods are often guided by trial and error, these simple cut-offs are subject to a certain degree of discretion as well as some local knowledge of the place under analysis (McMillen, 2001, p. 449). However, it must also be noted that a certain degree of discretion is used in almost every method of identification, even in econometric ones, where certain statistical confidence levels have to be chosen.

Another methodology based on simple cut-offs is that employed by Shearmur and Coffey (2002), which – following the idea by McDonald (1987) – identifies as sub-centres the contiguous TAZs that show an employment to resident workers ratio (*E/R* ratio) greater than 1 and a number of jobs of at least 5000 units. This method should be actually classified somewhere in between static and dynamic approaches since it approximates the net in-commuting even if it does it by using stock data. The *E/R* ratio is preferred over employment density by some authors because the former approximates the capacity of attraction of employment more accurately (Shearmur & Coffey, 2002, p. 579).¹ However, it puts less emphasis on employment localisation. In fact, Giuliano and Small (1991, p. 167) argue that *E/R* ratio does not sufficiently take into account agglomeration economies, which depend on the distance between firms (density) rather than on the presence of resident workers.

A third methodology is based on the identification of local density peaks. Following this approach, the zones which present a local maximum in employment density or in the *E/R* ratio, with respect to the surrounding area, can be considered sub-centres. Following the idea by Bender (1981, 1985) on the effects of sub-centres on land prices, McDonald (1987) identifies sub-centres by looking at local peaks in the *E/R* ratio. Similar approaches can be found in McDonald and McMillen (1990), while Gordon, Richardson, and Wong (1986), Craig and Ng (2001) and Redfearn (2007), among others, look at local peaks in employment density. In the

¹ For a more detailed analysis of the advantages and disadvantages of the *E/R* ratio for the identification of sub-centres, see Coffey and Shearmur (2002) and Forstall and Greene (1997).

last two studies, authors identify local peaks using a non-parametric specification of the density function in order to avoid what in their opinion are excessively strong assumptions regarding the spatial distribution of sub-centres. In fact, in most cases, the analysis starts from the assumption of a monocentric structure (Mills, 1972; Muth, 1969), estimating a simple negative exponential density function (Kemper & Schmenner, 1974). Another possibility is the assumption of a polycentric urban structure adopting a parametric density function with multiple centres. In this respect, Heikkilä et al. (1989) present three explicit alternatives to specify the polycentric density function on the basis of the value given to the access to centres.

A fourth method relies on the identification of the residues of an econometric estimation of the density function, which can be parametric or non-parametric, depending on the strength of the assumptions on the urban structure. The residues that are statistically diverse from zero can be considered candidates for sub-centres (McDonald and Prater, 1994; McMillen, 2001). These selected nodes are then analysed in terms of their effects on urban spatial structure through the estimation of a polycentric model of employment density. A recent work by Leslie (2010) identifies Phoenix sub-centres with a Kernel smoothing process for both employment and establishment density. Using Kernel density measures with point data has the advantage of allowing a minor bias in the spatial definition of centres due to administrative boundaries and land uses (Adolphson, 2009). Moreover, McMillen and Lester (2003) argue that econometric methods are more objective and work better for comparative analysis among different MAs. However, for the analysis of MAs over time, simple threshold methods are usually preferred.

A fifth morphological method uses spatial statistics tools to identify – within a MA – clusters of zones with higher density than that of the surrounding ones. High-density clusters can be considered sub-centres since they represent relative concentrations of employment. This approach is based on the Local Indexes of Spatial Autocorrelation (LISA) (Anselin, 1995). With these indicators, it is possible to quantify the degree of clustering of neighbouring zones with high levels of density (Riguelle, Thomas, & Verhetsel, 2007). This method has been used, among others, by Baumont, Ertur, and Le Gallo (2004) and by Guillain, Le Gallo, and Boiteux-Orain (2004).

Interaction methodologies

The functional approach based on interaction procedures seems particularly sound for European metropolitan areas, which are characterised by a number of small and historically determined urban centres that coalesced in larger and integrated areas (Calafati, 2002, 2009). During the first wave of the growth of city boundaries – when urbanisation was accelerating speedily – the expanding MAs absorbed the smaller centres nearby through the process of *absorption* (Champion, 2001). During the last fifty years, however, cities have continued to expand spatially, but at a slower rate than the expansion of their field of influence. Hence, MAs have grown by *integration* (Champion, 2001), which consists mostly in a relational concept rather than a physical one.

The majority of European MAs are the result of such a process of coalescence and are characterised by a dominant centre, which is surrounded by other smaller centres, forming contemporary polycentric MAs. Most European metropolitan sub-centres did not come about from a simple process of decentralisation from the city centre. They were pre-existent to the formation of MAs, and they still play the role of central places for at least a fraction of the metropolitan territory even without any particularly high employment density. Theoretically, the most important framework for the interpretation of sub-centring in European MAs is the Central Place

theory (Christaller, 1966; Lösch, 1954) since it focuses on urban functions and hierarchies between territorial nodes. From this perspective, metropolitan sub-centres can be seen as central places within a given MA, places that functionally organise their surrounding territory. However, since the Nineties, this theoretical framework has been partially re-thought in terms of horizontal relations between centres, rather than hierarchical ones, giving rise to the network of cities paradigm (Batten, 1995; Camagni & Salone, 1993; Dematteis, 1990). The underlying idea is that different centres specialise in different sectors in order to benefit from Marshallian economies of location and relate to one another in a non-hierarchical way. This theoretical framework is particularly sound for the study of urban systems and inter-metropolitan links, while for single MAs – as is the case in this work – the traditional central place theory can still play an important role.

The literature on the identification of metropolitan sub-centres with a functional approach includes some interaction methods which make use of flow data. This kind of studies can be found in both America (Bourne, 1989; Gordon, Richardson, & Giuliano, 1989) and in Europe (Burns, Boix, & Roca, 2001; Clark and Kluijpers-Linde, 1994). Bourne (1989) visually analyses commuting flows for Canadian cities. Gordon et al. (1989), on the other hand, follow a trip-generation density approach for Los Angeles, analysing the distribution of traffic flows among TAZs. In Europe, Burns et al. (2001) identify sub-centres in Spanish MAs by selecting the municipalities that show a net entry flow of at least 15%.

More recently, these interaction approaches have also been used for the analysis of spatial configuration at the level of national urban systems (Limtanakool, Dijst, & Schwanen, 2007; Limtanakool, Schwanen, & Dijst, 2009) or even at global level (Derudder et al., 2003). More specifically, former studies propose a set of spatial interaction indices, trying to quantify the strength and symmetry of interaction and the structure of the whole network (urban system), by using commuting flows. In few other works, commuting flows have been used to investigate the interdependence of suburbs and central cities within MAs (Barbonne, Villeneuve, & Thériault, 2007; Gilli, 2002; Holly, 1993). In particular, Holly (1993, p. 419) argues that commuting approximates the actual interaction between cities (exchange of labour, thus of knowledge and information), making clear the different position of each city in the metropolitan hierarchy.

Interaction methods are deeply different from those based on employment density. In fact, underlying these two alternative perspectives are two different concepts of metropolitan centre and, consequently, two different concepts of polycentricity. Gordon and Richardson (1996) argue that with dynamic methodologies it is possible to grasp the role of sub-centres not only as employment concentrations, but also as focal points of a MA in terms of the urban functions that sub-centres supply to their neighbouring territory. This standpoint makes it possible to shift from a morphological approach – where the major focus is on agglomerations of activities in a given built up area (European Foundation, 1998, p. 60) – to a functional one, where sub-centres are conceived as places that have the capacity to generate a metropolitan structure (Roca Cladera, Marmolejo Duarte, & Moix, 2009, p. 2860).

On the whole, morphological and functional approaches are substantially different. However, they are not necessarily in contradiction to each other. In this respect, a comprehensive ESPON study on polycentricity measures the spatial structure of European functional urban areas both in terms morphological and relational perspectives (Nordregio, 2005). Authors consider three dimensions in their analysis, namely size, location and connectivity, where the former two are morphological and the latter is a functional aspect. More recently, Adolphson (2010) describes urban structures in the Stockholm city region by combining spatial-density modelling

using kernel convolution with polycentric density estimations and mixed functionality.

Methods and data

This section proposes a methodology for the identification of metropolitan sub-centres that combines interaction and pure functional elements. Before starting to describe this methodology, it is important to discuss the units of analysis. Polycentricity is a multi-scalar concept that can be studied at three different spatial levels: country, inter-urban/regional and urban level (Davoudi, 2003). In this work, the latter level of analysis has been chosen and sub-centres have been identified and selected among all the urban nodes located within each MA. Since Italy still lacks an official definition of MAs, the latter have been taken from a recent work by Boix and Veneri (2009) – which provide the identification of all Italian MAs by adopting an algorithm inspired by the 1990 methodology of the US Federal Register (Office of Management & Budget, 1990).

The procedure starts from the conceptualisation of MAs as networks of cities (municipalities). Networks are made up of two main components – urban nodes (municipalities) and the links between each node (commuting flows). Defining MAs as networks of cities – or as cities of cities (Roca Cladera et al., 2009) – allows us to use concepts and tools of systems theory to analyse them (Boix, 2002). Moreover, most of the network analysis indicators can be usefully employed to mathematically quantify the relations between urban nodes within a given MA. The idea consists of studying the centrality structure of MAs, concentrating on the way each urban node relates to the others. Hence, central places can be identified by analysing the way relationships are distributed within MAs, rather than at the static characteristics of the territory.

Using data on flow instead of data on stock can allow this. Going into detail, many indicators of centrality can be borrowed from the network analysis in order to quantify how each node plays a central role in the network. The most simple, robust and easy-to-interpret indicator is the degree of centrality (Freeman, 1979) which, in asymmetrical networks, – such as MAs – can be approximated by node degree (Zemljčić & Hlebec, 2005, p. 82). The in-degree index represents the number of links that directly enter each node. At the same time, the out-degree is the number of links that directly exit each node. Hence, in order to detect those municipalities that exert a central role in a metropolitan network, a flow-centrality ratio (FC_i) has been computed based on the following formula:

$$FC_i = \sum_{j=1}^N l_{ij} / \sum_{i=1}^N z_{ji},$$

where the numerator and the denominator are, the *in-degree* and the *out-degree* indicators for the i th municipality of a given MA, respectively; l_{ij} is the number of trips towards the municipality i from the j th municipality of the MA under analysis. At the same time, z_{ji} is the number of trips from the i th municipality towards the other j th municipalities of the same MA. The centrality of each node is directly proportional to the numerator and inversely proportional to the denominator. This occurs because a municipality that captures most of the flows reveals its capacity to dominate its surrounding territory from a functional point of view.

Another indicator that has been used in the procedure for the identification of sub-centres is the directional dominance index (DII_i), which has been introduced by Limtanakool et al. (2007). This indicator measures the degree of involvement of a node in the network and can be calculated as follows:

$$DII_i = I_i / \left(\sum_{j=1}^J I_j / J \right),$$

where I_i is the number of inward journeys to node i th from all the other j nodes of the same MA; J is the number of nodes in each MA, for $i \neq j$. Differently from the FC_i ratio, which is a pure centrality index, DII_i allows taking into account the more general concept of power within each metropolitan network. In fact, this indicator depends on the total number of links and also takes into account, although indirectly, the size of each node.

The FC_i and the DII_i ratios have been calculated for each municipality of the two most important Italian MAs: Rome and Milan. These ratios have been calculated by using 2001 data on commuting flows between municipalities, provided by the Italian National Institute of Statistics (Istat). For each MA, there is a matrix of commuting flows which contains the number of commuters from every municipality to all the others of the same MA.

The procedure of sub-centre identification proposed in this work has been developed following three steps. Firstly, the FC_i and DII_i ratios have been calculated for all municipalities within each MA. Then, all the municipalities with FC_i and DII_i ratios higher than 1 have been considered as candidate sub-centres. Initially, these selected nodes are considered only “second order” sub-centres because work-commuting flows do not represent all the movements that take place in MAs, shaping the hierarchies of their sub-areas. Urban nodes can indeed be relevant – or central – also for activities related to consumption, study and leisure. As a consequence, it is necessary to distinguish the concept of employment sub-centre from the wider one of urban sub-centre. While the former can be correctly identified using the in-degree indicator alone, the latter requires some other measure to grasp the actual capacity of sub-centres to organise their surrounding territory by providing a wider set of urban functions.

Therefore, as a second step, another indicator – called *productive completeness* (PC_i) – has been calculated for all municipalities of the two MAs. This indicator approximates the variety of functions supplied by each urban node, starting from the idea that besides attracting commuters, a metropolitan sub-centre must have a minimum degree of productive variety which can be thought of as a sign that a wide range of urban functions is supplied by a given node. The PC_i index can be calculated as follows:

$$PC_i = s_i / \hat{s},$$

where PC_i is the productive completeness for the i th municipality; s_i is the number of 5 digit sectors with at least one job in 2001, using Istat Census data with NACE sectorial classification;² $\hat{s}$ is a parameter indicating the average number of 5 digit sectors in each MA. The aim of this indicator is to grasp the urban character of candidate sub-centres by attempting to measure urbanisation economies (Jacobs, 1969) similar to what has been done by Compagnucci (2009). In other words, central nodes must show typical urban characteristics, such as completeness and diversity of their sectoral structure, and they must prove to be net suppliers of functions. Since the aim of this second step is that of catching the capacity of each municipality to provide the widest variety of functions, the relative number of sectors has been preferred to other measures of productive diversity, like the Hirschman–Herfindahl index. The latter is more appropriate for grasping the extent to which the employment of a given economic system is specialised in few sectors or evenly distributed in wider set of activities. However, it should be recognised that the number of 5 digit sectors represents only an approximation of the economic functions provided by each centre. The relationship between NACE classification and economic functions should be further investigated, but this is not the purpose of this work.

² In the Nace classification system, the 5-digit sector is a very detailed level of description. To give an example in the manufacturing industry, a 5-digit description is the reproduction of sound recordings.

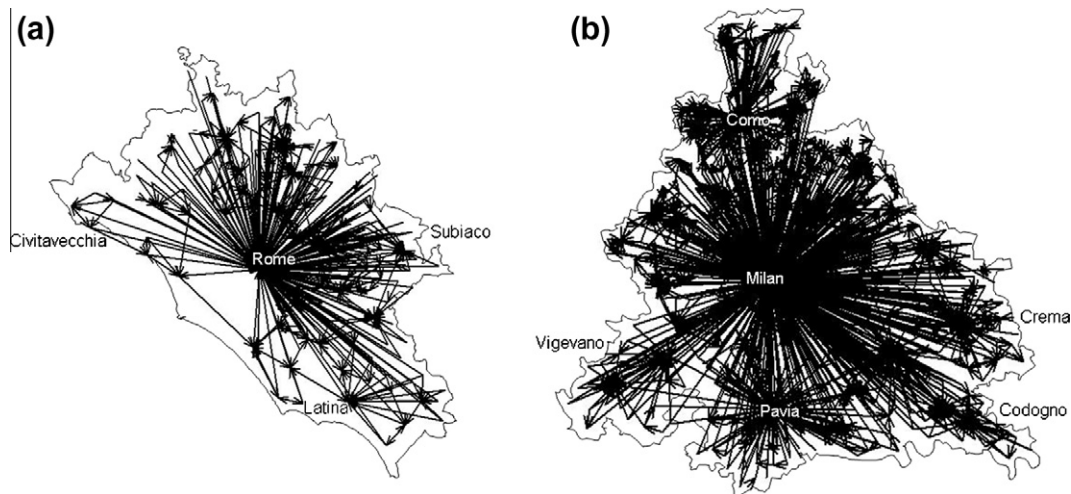


Fig. 1. Visual representation of links within the MA of Rome and Milan (flows of commuters for work reasons). Black lines represent commuting links among municipalities. (a) Rome. (b) Milan.

The third and last step of the procedure consists of selecting, for each MA, those municipalities for which both FC_i and DII_i ratios, as well as PC_i index are higher than 1. Those municipalities which satisfy all three criteria can be considered as first order sub-centres or, in other words, metropolitan sub-centres. They can be considered, within their respective MAs, as net dominators, net providers of functions and places with a positive net centrality. Moreover, this method also makes it possible to grasp a scale factor without imposing any absolute population or employment threshold. In fact, a larger dimension is associated with a higher value of both centrality (FC_i and DII_i) and completeness (PC_i).³ The results of the proposed methodology, as applied to Rome and Milan MAs, are reported in the next section.

Empirical findings

In this section the results of the above-mentioned procedure are reported and compared with those obtained using the traditional density-based method of Giuliano and Small (1991) (GS) as well as the E/R ratio method by Shearmur and Coffey (2002) (SC). In regards to the methodology proposed in this work, Fig. 1 visually represents the commuting flows among municipalities in the two selected MAs. In order to make the figure easier to read, each matrix has been dichotomised, where for each municipality – hence for each row of the matrix – there are four 1-values in correspondence of the four nodes with the highest interaction with the former (four-nearest-neighbour matrix). The figure highlights the typical polycentric structure of Milan, with several central nodes surrounding the pivot municipality. Each central node, corresponding to the thickening of links, is clearly visible. The FC_i ratio is adopted in order to understand how thick each node is, which in turn means how central each node is. While Milan has quite a large number of sub-centres all around the pivot municipality, Rome's spatial structure appears to be less polycentric, with few central nodes and with the pivot municipality having a prominent role. Both the polycentric and complex structure of Milan and the classic monocentric organisation of Rome are generally consistent with previous analyses that described the spatial structures of the two areas (Corda, 2003; Boatti, 2008). Basic descriptive statistics of the employed indicators are reported in Table 1 for each MA.

Table 1

Functional indicators: descriptive statistics.

	No. obs.	Min	Max	Mean	Std. dev.
Rome					
FC_i	200	0.02	4.62	0.38	0.51
DII_i	200	0.00	195.94	1.49	13.88
PC_i	200	0.02	2.02	0.33	0.31
Milan					
FC_i	597	0.00	4.63	0.58	0.47
DII_i	597	0.00	235.07	1.11	9.71
PC_i	597	0.01	2.01	0.37	0.28

Fig. 2 shows the municipalities that have been identified as metropolitan sub-centres in darker colour. These municipalities are characterised by a high internal attractiveness ($FC_i > 1$), a high dominance level ($DII_i > 1$; they are already 'second-order' sub-centres) and high productive completeness ($PC_i > 1$). Milan confirms its typical polycentric spatial structure, while the same thing cannot be said for Rome, which appears to have a much smaller number of sub-centres. The adopted methodology identifies four central municipalities for Rome, which have been consolidated in three sub-centres on the basis of contiguity and interaction criteria.⁴ In the case of Milan, 11 municipalities have been identified and aggregated in nine consolidated sub-centres. All the metropolitan sub-centres of the two MAs are listed in Table 2 where the number of municipalities composing each sub-centre is reported in brackets. Table A1 in "Appendix" shows the list of all municipalities that meet the requirements to be sub-centres.

Results obtained with the proposed approach can be compared with those emerging from the application of the traditional GS and SC methods. Giuliano and Small identify two thresholds beyond which a place can be considered a sub-centre. The first is the total number of jobs, which must be at least 10,000 units and the second is the employment density of at least 10 jobs per acre. It must be noted that the choice of thresholds is substantially arbitrary and those by Giuliano and Small have been set on the basis of Los Angeles' territorial structure. Moreover, they apply such thresholds to TAZs, which are much smaller units than Italian municipalities. In

³ The Pearson coefficient of correlation between each municipality's total employment and the three indicators used in this work – FC_i , DII_i and PC_i – are, respectively, 0.89, 0.97 and 0.36.

⁴ Selected municipalities which are contiguous have been consolidated in a single centre, which takes the name of the most populated municipality. The resulting spatial aggregate must show at least a level of self containment in terms of commuting flows of at least 15%.

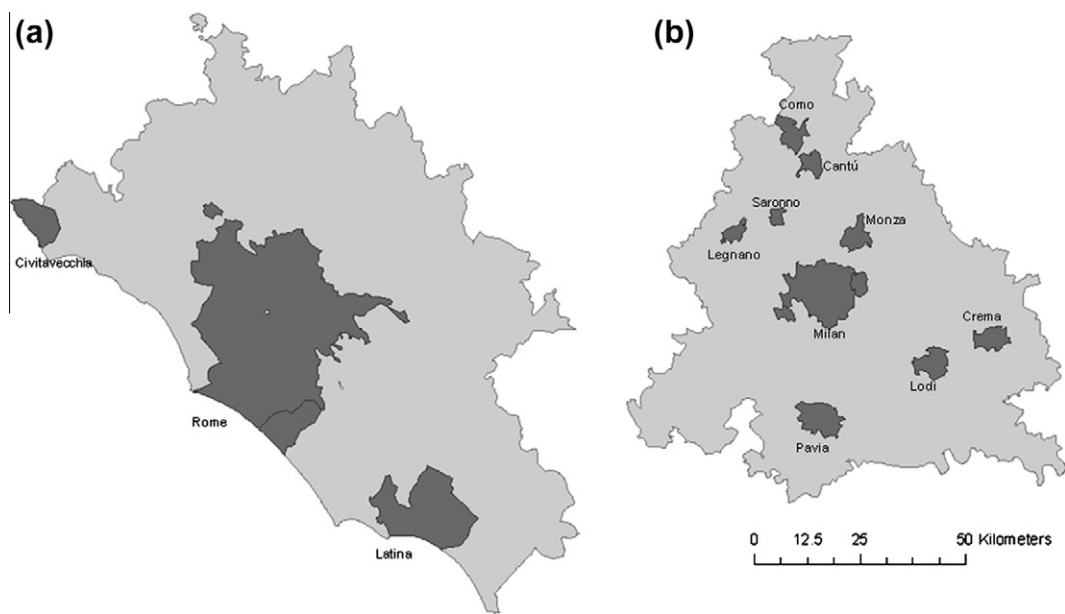


Fig. 2. Metropolitan sub-centres of Rome and Milan identified with the functional approach. (a) Rome. (b) Milan.

Table 2
List of metropolitan sub-centres in Milan and Rome MAs.

Method	Rome MA	Milan MA
Functional	Rome, Latina, Civitavecchia	Milan (3), Como (1), Pavia (1), Crema (1), Lodi (1), Saronno (1), Cantù (1), Legnano (1), Monza (1)
SC	Rome, Colleferro, Civita Castellana	Milan (19), Pavia (1), Lodi (1), Crema (1), Treviglio (1), Trezzo sull'Adda (1), Merate (1), Vimercate (3), Carate Brianza (1), Cantù (1), Como (1), Erba (1), Olgiate Comasco (1), Castellanza (1), Magenta (1)
GS	–	Milan (3)



Fig. 3. Sub-centres of Milan and Rome, identified with the Shearmur and Coffey's (2002) method.

fact, by using the GS method, no sub-centres were identified in the case of Rome. Not even did the CBD meet the thresholds, and only one sub-centre, composed of three contiguous municipalities including the CBD, was found for Milan. Given these results, an employment gross density of 10 jobs per acre seems to be a very

high threshold for Italian municipalities and can be very sensitive to the territorial units of analysis at which data are available. However, by adopting the SC method, based on the E/R ratio (higher than 1) and a minimum of 5000 jobs, 35 municipalities in the case of Milan and six in the case of Rome meet the necessary requirements for

selection. These have been aggregated in fifteen and three consolidated sub-centres, respectively (Table 2 and Fig. 3).

Evaluation of methods

Comparing the results of the three methodologies (Table 2 and Figs. 2 and 3), it emerges that the GS procedure is not able to identify sub-centres in the Italian metropolitan context, especially in the case of Rome where no municipalities have been selected. However, the SC method allows for the identification of the highest number of sub-centres. Probably, by simply adjusting employment thresholds, it would be possible to obtain very similar results to those obtained with the functional method proposed in this work. However, a more relevant difference between the two approaches can be noticed clearly by observing the maps (Figs. 2 and 3). Looking at Milan, for example, it appears that with the SC approach almost all the pivot's first belt municipalities – plus some of the second belt – are selected as sub-centres. This is a comprehensible result, since employment density tends to decrease from the pivot's peak towards the hinterland. Hence, municipalities surrounding central places can still provide a relevant number of jobs. However, considering those first and second belt municipalities directly as sub-centres does not seem to be appropriate since in some cases they are not proper central places, but just high-job-density nodes due to their proximity to the pivot. In fact, these nodes do not show a positive net centrality and, most importantly, they are not net suppliers of functions. They could be considered sub-centres only by conceptually aggregating them to their respective pivot municipalities in a single territorial unit, which in the case of related CBDs could be thought of as *de facto* cities (Calafati & Veneri, 2011), maybe too large if the aim is to identify just the centres. On the contrary, interactive approaches based on commuting, combined with a measure functions supplied (productive completeness), seem to be more restrictive, identifying fewer and smaller units and highlighting the places which are actually central and which organise their surrounding territory. This result can be reached without imposing absolute discretionary thresholds.

The qualitative and critical comparison made in the previous paragraph could be the most appropriate way to contrast methodologies. However, the analysis of density patterns can tell us the degree to which functional methodologies are consistent with traditional threshold-based approaches. In order to contrast the effectiveness with which each identification method is able to explain the spatial organisation of MAs in terms of density, results can be compared with the estimation of a polycentric density function, similar to what has been done by Anas et al. (1998), Roca Cladera et al. (2009) and García-López (2010). More specifically, in order to verify whether the identified sub-centres explain metropolitan spatial structure, two functional forms have been used for reasons of robustness. The first represents the most standard method and consists of estimating the Eq. (1), which is a log transformation of a standard negative exponential function, estimated with OLS:

$$\ln D(d_{CBD}, d_{SUB}) = \alpha + \beta \ln d_{CBD} + \sum_i \gamma_i \ln d_{SUB_i} + \varepsilon, \quad (1)$$

where D is the gross employment density per square kilometre in each municipality; d_{CBD} and d_{SUB_i} are the distance from CBD and from the i th consolidated sub-centre, respectively; α is the intercept; β and γ_i are the density gradients referred to the CBD and sub-centres, respectively; ε is the zero mean and constant variance error term.

In addition to the log of the standard exponential function, the logarithmic version of the gravitational model has also been estimated through the following equation:

Table 3

The adjusted R^2 values for the models contrasted.

	Rome		Milan		
	Functional	SC	Functional	SC	GS
Log exponential	0.525	0.393	0.671	0.674	0.426
Log gravitational	0.546	0.396	0.703	0.711	0.414

$$\ln D(d_{CBD}, d_{SUB}) = \alpha + \beta * d_{CBD} + \sum_i \gamma_i * d_{SUB_i} + \varepsilon, \quad (2)$$

where distances from CBD and from sub-centres are not log-transformed.

According to Roca Cladera et al. (2009, p. 2857), the gravitational model allows us to better grasp the role of CBD with respect to that of other sub-centres without underestimating the role of the former. Adjusted R^2 values of the models tested for both Rome and Milan are reported in Table 3. From this, it is possible to compare how much each identification method explains the metropolitan spatial structure.

Table 3 illustrates that in the case of Rome, the functional method proposed in this work allows for a much larger explanation of employment-density variance. In fact, by looking at both the standard exponential and the gravitational specifications, adjusted R^2 values are always much higher – at least 13.2% points higher – under the hypothesis that sub-centres are those identified with the proposed functional method. In the case of Milan, it can be noted that by considering the sub-centres identified with the GS procedure, adjusted R^2 values are much lower than those obtained by considering sub-centres identified with the functional and SC procedures. No matter what specification is used, the last two methods appear to substantially explain the same share of variance in the case of Milan. On the whole, it must be acknowledged that the outcomes of each method in terms of R^2 could be influenced by the data structure (e.g. the size and number of spatial units), the chosen functions (e.g. commuting and productive completeness) and the actual urban structure.

Concluding remarks

The methodology proposed in this work to identify sub-centres presents many advantages and some limits in its applicability. On the whole, by comparing the functional procedure proposed in this work with other two methods traditionally used in the literature, the former appears to work well from both a theoretical and an empirical point of view, showing a better fit when estimating a polycentric density function. Theoretically, a functional approach is more appropriate for studying the central places of Italian MAs, which, as in many other European countries, are the result of a coalescence process rather than a consequence of a simple decentralisation of employment from a congested CBD. Density measures are based on this latter idea. From an empirical point of view, the two procedures of identification do not show the same results. The functional approach allows the identification of the nodes that can be considered sub-centres in a more restricted sense, while with the SC procedure many municipalities surrounding the CBDs are also thought of as sub-centres and can be aggregated into a larger centre. However, the threshold method of GS has proved to be inadequate when applied to the Italian metropolitan context, particularly when the thresholds are not updated on the basis of local knowledge.

Regarding the limits of using functional approaches, the availability of data represents a serious consideration, since data on flows are usually not as widely available (nor frequently updated) as data on stock. A modifiable areal unit problem (MAUP) can

Table A.1

List of all municipalities included in metropolitan sub-centres of Milan and Rome MAs.

Method	Rome MA	Milan MA
Functional SC	Rome, Pomezia, Latina, Civitavecchia Rome, Pomezia, Frascati, Fiumicino, Collesferro, Civita Castellana	Milan, Como, Pavia, Crema, Lodi, Segrate, Saronno, Trezzano sul Naviglio, Cantù, Legnano, Monza Milan, Como, Pavia, Segrate, Lodi, Saronno, Crema, Cantù, Trezzano sul Naviglio, Vimercate, San Donato Milanese, Cernusco sul Naviglio, Agrate Brianza, Corsico, Rozzano, Assago, San Giuliano Milanese, Treviglio, Peschiera Borromeo, Arese Settimo Milanese, Lainate, Magenta, Erba, Castellanza, Carate Brianza, Merate, Vimodrone, Pero, Novate Milanese, Cassina de' Pecchi, Concorezzo, Caronno Pertusella, Olgiate Comasco, Trezzo sull'Adda
GS	–	Milan, Corsico, Sesto San Giovanni

emerge when applying these methods, especially in analyses across countries. In fact, the choice of the unit of analysis is heavily dependent on the spatial level at which functional data – e.g. commuting – are available, while for many density-based approaches these kind of problems are much less likely to arise and a more detailed spatial level is allowed. However, density-based approaches based on thresholds cannot be easily applied to different contexts, because of differences in the settlement structure and in the spatial units for which data are available.

On the whole, if we are interested in evaluating the extent within which metropolitan areas are polycentric by looking at the number of sub-centres (Riguelle et al., 2007), the choice of the method employed for the sub-centres identification can be very influential on the assessment of metropolitan spatial structures. In the end, the choice of the method should primarily depend on whether to look at morphological polycentricity or functional polycentricity. The difference is mainly in the concept of centre. In fact, employment size and density are good measures of centre nodality, hence of the absolute importance of a centre. On the other hand, the provision of functions and jobs in excess of the amount demanded by residents is a measure of centrality and of the relative importance of a centre with respect to its surrounding territory. This work has contributed to the literature on this latter approach.

Further developments should examine whether there are different kinds of sub-centres that differ in terms of their origin – decentralisation or coalescence. It is worth investigating whether different kinds of sub-centres have different impacts on a metropolitan spatial structure. This could be investigated by looking at the way sub-centres relate to one another and by observing the extent to which their functions are alternative or complementary. In order to do this, other connections should be explored, such as consumption and leisure activities and inter-firm exchanges.

Appendix A

See Table A.1.

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